

DEEP REINFORCEMENT LEARNING

365.382 (SS26)

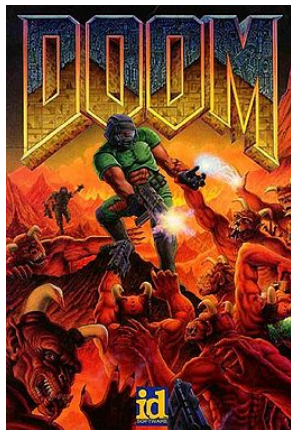
Assignment IV: Play DOOM with RL



LIT AI LAB
Institute for Machine Learning

Final Challenge

The final challenge is based on the **1993 FPS game DOOM** by id Software.



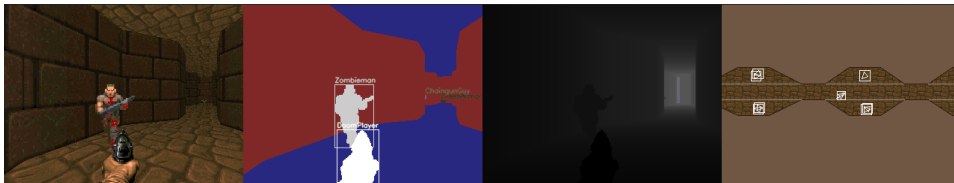
- Fast-paced "3D" first-person shooter.
- PvE / PvP game modes, three maps.
- **Grading:** PvE against (easy) bots, simple scoring.
- **Tournament:** PvP against fellow students.

In the final lecture we will have some tournament highlights.

VizdoomMPEnv

Environment available at <https://github.com/gerkone/jku.wad/tree/main>

- Basic reward included → **improvements are up to you!**
- Should run at 200-500FPS
- Partially configurable, on state space and basic game settings.
- Chose between RBG (screen) + extra game buffers: labels, depth, automap



DOOM against bots



- **States:** 128×128 game frames (+ buffers)
- **Actions:** discrete, 8 (7 game buttons + noop)
- **Rewards:** +hits, -damage, ++frags
- **Grading:** only depend on performance against bots.

Recommendations

- Basic DQN agent and simple reward are included
→ **improvements are up to you!**
- Learning from game frames → need a visual encoder.
- Agent observations can be configured freely!
- You are free to adapt the reward signal
→ **moving etc.**
- Visualize the behavior of your agent
- You can try whatever you like, be creative

Submission

- Submission server: <https://apps.ml.jku.at/challenge>
- Export your trained agent as ONNX file
- Upload ONNX file to submission server
 - ☐ Code submission closes on **Sunday, June 7th, 18:00**.
 - ☐ Max model file size: **50MB**.
- Submit your **code**, **best model** and a short **report** to Moodle
 - ☐ Report the performance on your local setting.
 - ☐ Moodle submission stays open 24h after the challenge closes.
- Please upload a zip file named `k<studentid>.zip` containing:
 - ☐ `code.ipynb` / `code.py`
 - ☐ `model.onnx`
 - ☐ `report.pdf`

Evaluation

Default "ROOM" environment, 4 bots

- $< 0 : 0$ Pts.
- $\geq 0 : 2$ Pts.
- $\geq 20 : 4$ Pts.
- $\geq 40 : 6$ Pts.
- $\geq 60 : 8$ Pts.
- $\geq 80 : 10$ Pts.
- $\geq 100 : 12$ Pts.
- $\geq 120 : 14$ Pts.

Presentations

- I'll ask students with the best scores to present.
- Students selected to present, will receive +2 bonus points.